

This Month's DVD Contents

This month's DVD contains various emulators, simulators and a selection of utilities that aid the electronics design and simulation process

SNEHA AMBASTHA

FreeRTOS

FreeRTOS is a real-time operating system (RTOS) that can run even on a small microcontroller (MCU), though is not limited to it. In special cases, as in MCUs, full implementation of an RTOS is not possible because of the size of the MCU and the nature of the end application it is attached to. FreeRTOS fulfils the requirement for such components in any application.

Bochs IA-32

This is an open source emulator for Intel x86 machines, particularly IA-32. It can emulate a number of processors like 386, 486, Pentium or Pentium Pro and even AMD64 (with or without streaming single instruction, multiple data (SIMD) extensions (SSE), SSE2, 3DNow and MMX) using the common input/output (I/O) and customised basic input/output system (BIOS).

Flowcode

Flowcode is a GUI software that allows a user to quickly and easily develop complex electronic and electromechanical systems. The graphical programming tool allows even those with little experience to develop complex electronic systems in minutes. It supports testing and debugging of user code, simulation in 3D, component creation, Arduino programming in addition to supporting a large number of PIC, dsPIC, ARM and AVR devices. It is available in student, educational and professional versions as single/ten users/50 users and campus licences.

Meep

Meep, or MEEP, which is officially called MIT electromagnetic equation propagation, is a free finite-difference time-domain (FDTD) simulation software developed for the modelling of

Popular resources

- 7Zip. Version v.15.05 beta (file archiver)
- Java Development Kit (JDK). Version 8.0.450.15 (development kit to create Java runtime environment for various applications)
- VLC. Version 2.2.1 (media player)
- Opera. Version 30.0.1835.59 (Web browser)
- Free Download Manager. Version 3.9.6 build 1549 RC (file download manager)
- Thunderbird. Version 38.0.1 (email application)
- Mozilla Firefox. Version 39.0 (Web browser)
- Apache Open Office. Version 4.1.1 (open office)
- Ubuntu. Version 15.04 (operating system)

electromagnetic systems. It is also a freeware developed by MIT, distributed under GNU GPL licence. Meep can simulate in any of the coordinates (cylindrical, 1D, 2D and 3D) and work portably on any Unix-like OS.

CircuitLogix

CircuitLogix allows you to convert your PC into an electronics laboratory, which would give you access to thousands of interconnected components and devices. This tool not only enables you to design the schematic and simulate it in one program but also gives a fast simulation experience. You can test your electronic circuit designs in all possible scenarios that may arise without worrying about the parts getting damaged.

eSketch PRO

eSketch Pro is an electronic sketching tool that captures schematic and simulates analogue circuits. It has been specially created for circuits that contain inductors, capacitors, op-amps and resistors. This tool can change the layout, component tolerance, component values and can identify the gain bandwidth in an op-amp.

CIDER

CIDER is a combination of the latest version of SPICE3 (version 3F.2) and DSIM, which is a C based device simulator. This combination gives it an

ability to simulate both mixed-level circuits and devices. This mixed-level simulation capability makes CIDER a good choice to test circuit performance and to identify the technical parameters used there with utmost accuracy.

IdealCircuit

idealCircuit is a free simulator for analogue circuits with true or ideal components. This ideal components compatibility makes evaluation and testing of designs quick, providing a detailed analysis of the output. It supports Windows 7 OS and has the same algorithm as NL5 circuit simulator. It has a very simple interface with fewer models and components.

Code::Blocks

Code::Blocks is an open source integrated development environment (IDE) for developing C/C++ projects (DirectX, console applications, OpenGL, GTK+ and QT). Its target environment resembles that of Eclipse and thus can be referred to as its mini version. It has a consistent look, feel and operation across all platforms and is compatible with Linux, Mac and Windows. You can enhance its functionalities by either installing plugins or creating codes.

G'MIC

GREYC's Magic for Image Computing (G'MIC) is an open source, full-featured framework for processing images. It provides different user interfaces (UIs) for converting, filtering, manipulating and visualising generic datasheets for images to different signal levels. The interfaces include command line interface gmic, plug-in gmic_gimp, Web service G'MIC, Qt based interface ZArt, a portable C++ image-processing library. ●

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